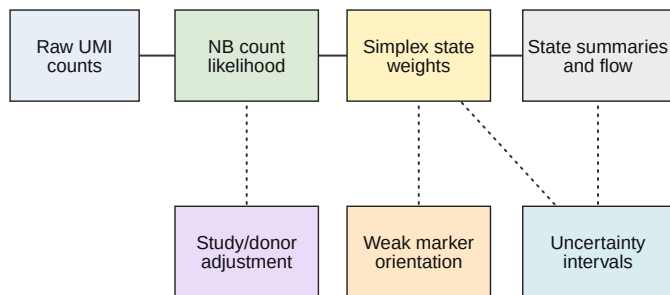


A Raw-count state-mixture model



B Model terms and outputs

Observation	Raw UMI counts enter the model with a library-size offset.
Likelihood	Negative-binomial means link counts to state-specific gene programs.
Latent state	Each cell carries continuous simplex weights rather than a forced hard class.
Adjustment	Study and donor terms absorb cohort- and sample-level count shifts.
Orientation	Marker sets orient states without replacing likelihood-based inference.
Readouts	Composition, FPI, transfer, uncertainty, and cross-sectional flow are reported as bounded outputs.

C Result-to-claim evidence spine

Fig. 2	Public raw-count run: NB fit, composition, FPI, and cross-sectional flow.	Feasibility demonstration, not a disease-mechanism result.
Fig. 3	Known-truth simulations: state-weight, dominant-state, entropy, transition, and marker recovery.	Computational recovery benchmark, not patient-tissue causality.
Fig. 4	Study-effect calibration: regularization tradeoff for cohort/donor adjustment.	Calibration readout, not universal hyperparameter proof.
Fig. 5	Transfer and bootstrap summaries: external state recovery with interval-aware reporting.	Transfer evidence with finite public cohorts.
Fig. 6	Cross-sectional flow summaries: state-linked disease shifts without causal timing claims.	Association-style flow, not longitudinal transition.
Supp.	Claim boundaries: model validation, uncertainty, and biological interpretation remain separated.	Defines what is supported now versus left open.

D Claim boundary contract

Supported now	Kept bounded	Not claimed
Raw-count NB state-mixture fitting; known-truth recovery; public data feasibility; transfer and uncertainty summaries.	Cross-sectional flow is descriptive; VI is reported as a posterior-mean skeleton; public cohorts constrain generalization.	No causal fibroblast transitions, no clinical biomarker, no therapeutic target proof, and no complete disease atlas claim.